Unit 1

Lesson 3

Spark The Challenge!

Engineers in Action!

Time Frame: [90 min]

Lesson Objectives:

By the end of this lesson students will be able to:

- Identify and describe the steps of the Engineering Design Process (Ask, Imagine, Plan, Create, Improve)
- Apply the "Ask" step to define the specific problem of designing an inclusive emergency alert system, considering user needs and constraints.
- Recognize relevant career paths in engineering, product design, and assistive technology.

Challenge questions of the lesson.

- What is the Engineering Design Process, and why is each step important when solving a problem?
- How can you use the "Ask" step to understand and define the problem of creating an inclusive emergency alert system?
- What are some potential career paths that involve engineering, product design, or assistive technology, and how do they contribute to solving real-world problems?

SEL Relation

- **Self-Awareness:** Recognizing the importance of keen observation skills, mirroring how sensors accurately collect data.
- **Self-Management:** Practicing precision and carefulness during experiments to ensure reliable observations and data collection.
- **Social Awareness:** Understanding how sensors are used to monitor environmental conditions and contribute to public safety and well-being (e.g., gas leak detectors, air quality monitors).
- **Relationship Skills:** Collaborating effectively during experimental setup and data observation, and openly sharing insights about how different sensors function.
- **Responsible Decision-Making:** Considering the ethical implications and importance of accuracy when collecting data via sensors, especially for safety-critical applications.

Engage

Teacher's Role:

- **Presenter**
  - o Introduce the career readiness video by linking it to the concepts explored in the previous lesson about the Engineering Design Process.
  - o Highlight how the skills and steps they've learned about designing a machine or product can be applied to real-world careers like product design, where the "Ask", "Imagine", and "Create" steps are essential in developing accessible solutions.
  - o Connection to Previous Lesson: Emphasize the connection between practical design work (like the DIY lab) and real-world applications. Reinforce how product designers use these steps to solve complex problems, such as improving accessibility.
- **Questioner**
  - o Ask reflective and open-ended questions to help students think about the practical aspects of designing inclusive products.

    "What do you think a product designer needs to know about user needs when creating a product for someone with a disability?"

    "How can the Engineering Design Process help product designers solve problems like accessibility challenges?"

    "What are the similarities between what we've done in class (designing machines) and what a product designer might do?"
  - o Connection to Previous Lesson: Use these questions to help students reflect on their previous understanding of the design process and apply it to the career-focused challenge of creating accessible products. Draw connections to the trial-and-error and improvement steps of the design process.
- **Supporter**
  - o Provide support as students work through the video and discussion, guiding them to make connections between the design concepts they've learned so far and the real-world design challenges that product designers face.
  - o Offer sentence starters like:

    "One challenge a product designer might face is…"

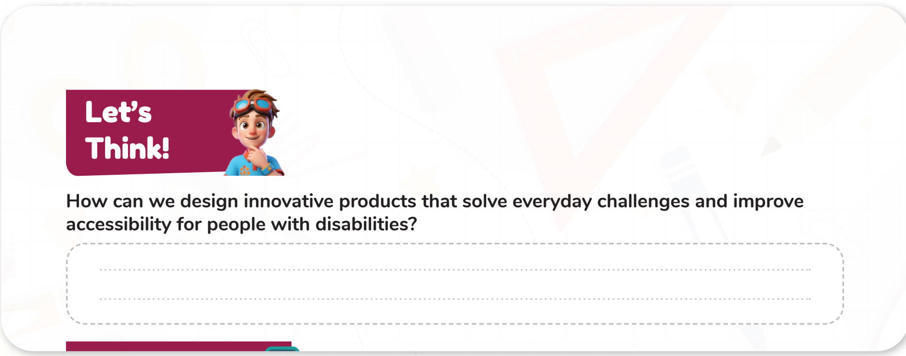
    "I think a product designer needs to understand…".
- **Collaborator**
  - o Participate in group discussions, listening to students as they share their insights and helping them connect their prior knowledge to new content. Offer encouragement and feedback as students collaborate to explore the role of a product designer.

- o Connection to Previous Lesson: Encourage students to collaborate and share ideas on designing accessible products, just as they did in the DIY lab activity. Reinforce the importance of peer feedback and iterative design in both the classroom activity and real-world product design.

Expected From Students:

- **Active Participation:**
  - o Students should actively engage in discussions by sharing their thoughts and ideas related to the "Let's Think" question about designing products for accessibility.
  - o They are expected to contribute their reflections on the video about product design careers and identify key challenges that product designers face in their daily work.
- **Critical Thinking and Reflection:**
  - o Students should reflect on the importance of designing inclusive products and be able to articulate why accessibility is a crucial consideration in product design.
  - o They are expected to make connections between the concepts presented in the video and the earlier class discussion about real-world design challenges.
- **Collaboration:**
  - o In small group discussions or pair work, students should collaborate to brainstorm solutions for improving accessibility through product design. They should demonstrate the ability to listen to others' ideas, build on them, and share their own insights.
- **Engagement with Career Readiness:**
  - o Students should engage thoughtfully with the career readiness discussion question by identifying key skills and challenges faced by product designers. They should demonstrate an understanding of the interdisciplinary nature of the field, connecting design principles with real-world applications.
- **Creativity and Problem-Solving:**
  - o During the Quick Pair Activity (optional), students should demonstrate creativity and problem-solving skills by brainstorming innovative product ideas that address accessibility challenges. They should show the ability to think critically about the needs of different users and how design can meet those needs.

Parts of the book included.



Let's Think!

Objective:

How can we design innovative products that solve everyday challenges and improve accessibility for people with disabilities?

Possible answers:

"We can create smart devices and tools that make daily tasks easier, such as voice-controlled home appliances, adjustable-height furniture, and wearable navigation aids for the visually impaired. By using technology like sensors, apps, and AI, we can help people with disabilities live more independently and comfortably."

Possible Strategies for Implementation:

1. **Pre-Discussion:**
   - **Strategy: Think-Pair-Share**
     - How to Implement:
       1. Ask students to individually think about challenges they've seen or experienced related to disabilities (e.g., difficulty accessing public spaces, using technology, or performing daily tasks).
       2. Pair students up to discuss possible solutions.
       3. Have students share their ideas with the class, creating a list of challenges and ideas on the board to refer to during the lesson.
2. **During Discussion: Active Engagement**
   - **Strategy: Brainstorming and Idea Mapping**
     - How to Implement:
       1. Write the "Let's Think" question on the board: "How can we design innovative products that solve everyday challenges and improve accessibility for people with disabilities?"
       2. Encourage students to brainstorm as many ideas as possible, either individually or in small groups, and map out their thoughts using mind maps or sticky notes.
       3. After brainstorming, have each group present one of their best ideas to the class, explaining why it would be beneficial for people with disabilities.
   - **Strategy: Round Robin**
     - How to Implement:
       1. In a Round Robin format, ask students to share one innovative product design idea they think could address an everyday challenge for people with disabilities.
       2. Each student will take turns speaking, and no idea can be repeated until everyone has shared one idea.

3. After the Round Robin, guide a class discussion on the potential impact of

Career Readiness

Objective:

Students will understand the challenges faced by product designers on a daily basis and the essential skills needed to succeed in the role. They will reflect on how these challenges connect to the design process they are learning about.

Possible Answers:

- **Challenges Faced by Product Designers:**
  - Balancing user needs with practical constraints (budget, time, materials, etc.).
  - Meeting deadlines while maintaining quality and innovation.
  - Collaborating with different teams (engineering, marketing, etc.) to create a product that works across multiple disciplines.
  - Conducting user testing and revising designs based on feedback or unexpected outcomes.
  - Navigating design iterations and addressing unforeseen issues or failures in the product design.
- **Skills Essential for Success:**
  - Problem-Solving: The ability to find innovative solutions to complex design challenges.
  - Creativity and Innovation: Constantly thinking outside the box to develop new and functional designs.
  - Collaboration: Working well with a team, including engineers, marketers, and manufacturers, to bring a product to life.
  - Technical Knowledge: A solid understanding of materials, tools, and technology to create functional and feasible products.
  - Attention to Detail: Ensuring every aspect of the design meets high-quality standards and addresses the needs of the users.

Possible Strategies for Implementation:

1. **Pre-Watching: Set a Purpose**

- **Strategy: Think-Pair-Share**
  - How to Implement:
    1. Ask students to think individually about the kinds of challenges a product designer might face (such as time constraints or meeting user needs).
    2. Pair students to share their ideas before watching the video.
    3. Discuss as a class, creating a list of challenges they expect to learn more about in the video.

1. **During Watching: Active Engagement**
   - **Strategy: Highlight & Code**
     - How to Implement:
       1. While watching the video, have students highlight key challenges mentioned by the product designer in the video.
       2. Encourage students to code their notes:
          - C for challenges faced
          - S for skills essential to the role
       3. Pause the video at strategic points to let students discuss what they've noted.
2. **Post-Watching: Discussion and Reflection**
   - **Strategy: Round Robin**
     - How to Implement:
       1. After watching the video, have students share one challenge and one skill they think is most important for product designers.
       2. Go around the room, with each student contributing a new idea. Encourage everyone to listen actively and build on others' answers.
       3. Conclude with a class discussion on how these challenges and skills relate to the Engineering Design Process they've been studying.

Explain

Teacher's Role:

- **Facilitator**
  - Guide students as they watch the video, prompting them to pay attention to the different steps of the Engineering Design Process (EDP).
  - Encourage students to think critically about how engineers approach design challenges and how each step contributes to developing a solution.
  - Pause the video at key moments to reinforce important concepts and ask questions like:

    "What step in the Engineering Design Process is the engineer focusing on right now?"

    "How does the engineer define the problem before jumping into the solution?"
  - Support students in understanding the application of the process by connecting the video content to the lesson and their own project work.
- **Questioner:**

Ask open-ended questions during the video to spark critical thinking:

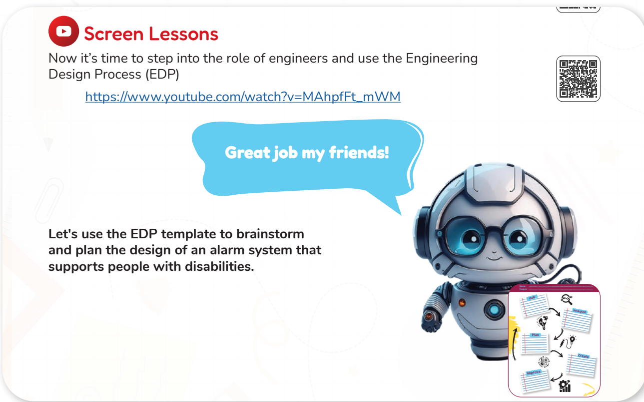
- “What might happen if one of these steps were skipped?”
- “Can you think of any examples where this process has been used in the real world?”

- **Presenter**
  - Introduce the video by explaining that engineers use the EDP to solve problems and create new products. Explain that students will follow the same process in the upcoming activity.
  - Link the video content to the Engineering Design Process they've already started to explore, emphasizing how the steps (Ask, Imagine, Plan, Create, Improve) help engineers build successful solutions.
- **Supporter**
  - Provide support during the video by guiding students to focus on specific aspects of the process. Offer clarification if needed, ensuring they understand each step and its importance.
  - Allow students to ask questions during the video and offer additional explanations or examples based on their queries.

Expected From Students:

- **Engagement:** Students should actively watch the video and take notes on the Engineering Design Process steps shown. They should be able to identify key phases, such as defining the problem, brainstorming ideas, creating prototypes, and testing solutions.
- **Understanding:** Students are expected to demonstrate an understanding of each step in the EDP and how it applies to real-world design challenges.
- **Critical Thinking:** During the video, students should think about how they would approach a similar design challenge. They should reflect on the importance of problem-solving and iteration within the design process.
- **Reflection:** After the video, students should be able to explain the Engineering Design Process and discuss how each step contributes to successful product development. They will apply these insights to their own design projects in the next activity.
- **Collaboration:** If working in groups, students will discuss their observations and thoughts on the EDP and prepare to apply the steps to their own projects.

Parts of the book included.



Screen Lessons

Objective:

Students will understand the Engineering Design Process (EDP) by watching a video that demonstrates how engineers use a structured approach to solve real-world problems. They will reflect on the importance of each step in the process and apply this understanding to their own design challenges.

Video link: https://www.youtube.com/watch?v=MAhpfFt_mWM

Possible Strategies for Implementation:

1. **Pre-Watching: Set a Purpose**

- Strategy: Pre-Video Questioning

Implementation:

1. Before watching the video, ask students the following questions to get them thinking about the Engineering Design Process (EDP):

   o "What do you think engineers do when they are asked to solve a problem?"

   o "Why do you think engineers need a process to help them design products or solutions?"

2. Have students share their ideas with a partner or in small groups, and write key thoughts on the board.

3. Set the purpose for watching the video by telling students they will be seeing how engineers use a process to design solutions and solve problems.

2. **During Watching: Active Engagement**

- Strategy: Focused Observation

Implementation:

1. Ask students to focus on specific steps of the EDP as they watch the video. Provide them with guiding questions, such as:

   o "Can you identify which step the engineer is working on right now?"

   o "What tools or methods is the engineer using to solve the problem?"

2. Encourage students to take notes on the different steps of the process and how the engineer addresses the problem.

3. Optionally, pause the video at key moments to allow students to discuss what they've seen and ask questions about the process.

3. **Post-Watching: Discussion and Reflection**

- Strategy: Collaborative Recall & Connection

Implementation:

1. After the video, facilitate a class discussion about the EDP.

   o "What steps did the engineer follow in the video?"

- "How did following this process help the engineer solve the problem?"

2. Have students work in small groups to recall the steps of the EDP they observed in the video, and discuss how they would apply these steps to a project of their own.
3. Have each group share one key insight or idea from their discussion. Encourage students to connect the process shown in the video to their own experiences with problem-solving in class.

Engineering Design Process

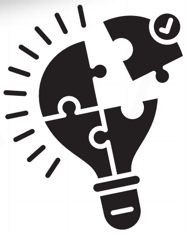
Name: ....

Project: ....

Ask

What types of disabilities should the alarm system accommodate (e.g., hearing, vision, mobility impairments)?

What are the unique challenges faced by people with disabilities when responding to traditional alarm systems?

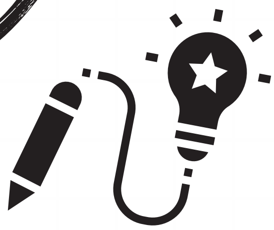


Imagine

Vibration-based alert system: Wearable devices that vibrate when the alarm is triggered.

Flashing lights or color-coded signals: For people with hearing impairments. Voice-activated alarms: Allowing people with mobility impairments to activate or cancel the alarm using voice commands.

Smartphone notifications: Sending a message or alert to the user's phone when the alarm goes off.



Plan

Materials Needed: What materials, tools, or technologies are required to build the system?

Design Specifications: What features should the alarm system include to make it accessible (e.g., vibration strength, visual indicators, sound volume)?

Technology Requirements: Will the system require programming or smart devices, such as sensors, alarms, or mobile apps?

Create

Improve

Elaborate and Evaluate

Teacher's Role:

- **Facilitator**
  - Guiding Design Thinking: Help students think through the process of designing an accessible alert device. Encourage them to consider real-world problems and how their device can solve them.
- **Encouraging Creativity**
  - Foster an environment where students feel comfortable expressing their ideas and making initial sketches, even if they're not perfect. Reassure them that the sketch is just a starting point, and creativity is key.
- **Supporter**
  - Move around the room, offering guidance as students sketch their ideas. Prompt them with guiding questions like:

    "How does your device meet the needs of someone with hearing impairments?"

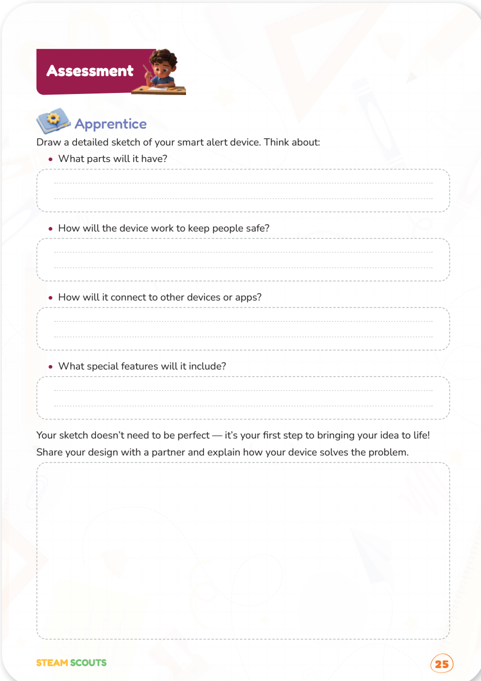
    "What materials could you use to make your device durable and comfortable to wear?"
- **Collaborator**
  - Encourage peer collaboration by having students share and explain their designs to partners. Support them in offering constructive feedback to each other.
- **Assessor**
  - Observe students' designs to gauge their understanding of accessibility and their ability to apply the Engineering Design Process to solve a real-world problem. Take notes on how they describe their devices and connect them to the problem they are solving.

Expected from students:

- **Design a Smart Alert Device:**
  - Draw a detailed sketch of your device, focusing on how it will help people with disabilities. Your sketch should include the parts your device will have, how it will work to keep people safe, how it will connect to other devices or apps, and any special features it will include.
- **Think Through the Design:**
  - Consider the functional elements of your device (buttons, vibrations, lights, etc.) and how these elements serve the needs of the user.
  - Explain why each feature is important for making the device effective and accessible for people with disabilities.

- **Collaboration:**
  - o After completing your sketch, share your design with a partner. Explain your device's functionality and how it solves a real-world problem.
  - o Provide feedback to your partner on their design and how it might be improved, considering accessibility needs.
- **Reflection:**
  - o Reflect on the challenges you faced during the design process and how you might improve your device in the future.
  - o Consider how your design can be tested and refined in real-world scenarios to meet users' needs.

Parts of the book Included.



Assessment

Apprentice

Draw a detailed sketch of your smart alert device. Think about:

- What parts will it have?
- How will the device work to keep people safe?
- How will it connect to other devices or apps?
- What special features will it include?

Your sketch doesn't need to be perfect — it's your first step to bringing your idea to life!
Share your design with a partner and explain how your device solves the problem.

STEAM SCOUTS 25

Apprentice

Objective:

Students will be able to imagine and visually represent a smart alert device by creating a detailed sketch that highlights its components, functions, and safety features, preparing them for the next steps of prototyping and testing.

Possible answers:

1. **What parts will it have?**

- **Vibration Motor:** For users with hearing impairments, a wearable vibration motor can alert them to the alarm.
- **LED Lights:** Bright flashing lights for visual alert, useful for users who are deaf or hard of hearing.
- **Speakers:** For users who are visually impaired, the device can emit sound at different frequencies or volumes.

- **Mobile App Integration:** An app to send alerts to the user's smartphone or other connected devices.
- **Button or Touch Sensors:** A simple button or touch sensor for the user to activate or deactivate the alarm.
- **Battery or Charging Unit:** To power the device for continuous use.
- **Bluetooth/Wi-Fi Module:** For wireless communication between the device and other smart systems or apps.

2. **How will the device work to keep people safe?**
    - **Multiple Alert Methods:** The device will combine vibration, light, and sound to ensure that the user receives the alert through one of the methods that works best for their **needs.**
    - **Real-time Notifications:** If connected to a smartphone app, the user will receive immediate notifications about the alarm and its source.
    - **Customizable Sensitivity:** The device will allow users to adjust the strength of the vibration or brightness of the lights to match their preferences.
    - **Emergency Response Integration:** The device could potentially connect to local emergency services or pre-set contacts (family, caregivers) to send an alert in case of an emergency.
3. **How will the device connect to other devices or apps?**
    - **Bluetooth:** The device can pair with smartphones, tablets, or smart home systems (like Amazon Alexa or Google Home) via Bluetooth for notifications and control.
    - **Wi-Fi:** The device can connect to the internet, allowing it to send alerts to smartphones and other devices remotely, even when the user is not within range of the physical alert system.
    - **Mobile App:** A dedicated app for Android or iOS could allow the user to customize settings, monitor battery life, and receive notifications on their mobile device.
    - **Smart Home Integration:** The device can trigger other smart home devices, such as turning on lights or activating a doorbell when the alarm is triggered.
4. **What special features will it include?**
    - **Voice Control:** For users with mobility impairments, the device could be voice-activated to turn the alarm on or off.
    - **Emergency SOS Button:** A large, easy-to-press emergency button that immediately notifies family or caregivers if the user needs help.
    - **Customizable Alerts:** Users can set different types of alarms for different emergency situations (fire, carbon monoxide, intruder) with corresponding sound patterns, vibrations, and light colors.
    - **Accessibility Features:** Text-to-speech for users with visual impairments to receive spoken alerts about the status of the system.
    - **Backup Power:** A backup power source (e.g., rechargeable battery) that keeps the system operational during power outages.

Possible Strategies for Implementation:

- **Modeling:** Show sample sketches or simple prototypes to inspire student ideas.
- **Think Aloud:** Verbally walk through the process of sketching your own device as an example.
- **Graphic Organizers:** Provide a template with sections for components, functions, and features to help organize sketches.
- **Pair Sharing:** Organize students into pairs or small groups to share and explain their sketches, encouraging constructive feedback.
- **Guided Questions:** Use prompts such as "What sensors will your device have?", "How will it alert users?", and "What makes your design unique?" to focus student thinking.
- **Encourage Iteration:** Remind students that sketches are a first step and they can update or improve their ideas later

Now I Can

Objective:

Students will reflect on their learning progress by reviewing the "Now I Can" statements and self-assessing their understanding and skills related to the Engineering Design Process, problem definition, brainstorming, and career awareness in engineering and product design.

Activity Instructions:

1. **Introduce the Purpose:**

Explain to students that this activity helps them evaluate what they have learned and identify areas they can improve.

2. **Review the Statements:**

Read each "Now I Can" statement aloud and clarify any unfamiliar terms or concepts.

3. **Self-Assessment:**

Ask students to rate their confidence level for each statement (e.g., "Confident," "Somewhat Confident," "Need More Practice") or use a simple scale (1–3).

4. **Evidence Sharing:**

Encourage students to give an example from class activities or projects that shows how they met each statement.

5. **Goal Setting:**

Have students note one skill or concept they want to practice more in future lessons.

6. **Discussion (Optional):**

Allow volunteers to share reflections with the class or in small groups to promote peer learning.

Relation to Standard:

1. ISTE standards
   - **ISTE Standard 1: Empowered Learner**
     - Students take an active role in their learning by setting goals, reflecting on progress, and making adjustments to improve.
   - **ISTE Standard 6: Creative Communicator**
     - Students communicate clearly and creatively using a variety of digital tools and formats.
2. NGSS (Next Generation Science Standards)
   - **NGSS Science and Engineering Practice 1:** Asking Questions and Defining Problems
     - Description: Students ask questions to investigate problems and explore solutions based on their observations.
   - **NGSS Science and Engineering Practice 7:** Engaging in Argument from Evidence
     - Description: Students construct and defend arguments based on evidence collected through investigations and experimentation.
3. Common Core State Standards (CCSS) for English Language Arts
   - **CCSS.ELA-LITERACY.SL.8.1:** Comprehension and Collaboration
     - Students engage in collaborative discussions, building on others' ideas and expressing their own ideas clearly.
   - **CCSS.ELA-LITERACY.W.8.10:** Write Routinely
     - Students write routinely for a range of tasks, purposes, and audiences.
4. CSTA (Computer Science Teachers Association) Standards
   - **CSTA Standard 1A-AP-08:** Design a Solution to a Problem
     - Students design solutions to real-world problems, applying principles of engineering and computational thinking.